Band-1 Hadamard Order 5476 — Miyamoto (Paley-I H_1368 + C_1370 over GF(37^2)); Miyamoto-only bonus NOT in SPK; Gap 2052 Still OPEN

H_684/H_1368/C_1370/H_5476 PASS; Paley-I/II H_5476 SKIP; 5476 Miyamoto-only NOT in SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_5444_Miyamoto`; `2026-09-15_Band1_Hadamard_5284_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_5476_miyamoto/`.
Cutoff. 2026-09-15T20:43:28Z (~4:43PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 684, 1368 and 5476 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with preferred Paley-I seed H_{1368} = Paley-I($q = 1367$) and alternate Double(Paley-I) seed (H_{684} = Paley-I($q = 683$); H_{1368} = Sylvester-double(H_{684})), with symmetric Paley conference C_{1370} over $\mathbf{GF(37^2)}$ ($q = 1369 = 37^2 \equiv 1 \pmod 4$, not a prime). Paley-II path for H_{1368} is **FAIL** ($q = 683 \equiv 3 \pmod 4$; needs $\equiv 1$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 5476 is a **Miyamoto-only bonus NOT in SPK**: Paley-I FAIL ($q = 5475 = 3 \cdot 1825$), Paley-II FAIL ($q = 2737$ not prime/power), Kronecker FAIL (4×1369). Same honesty pattern as 5284/5156/4916 (Miyamoto bonus). `clay_closed = false`.

Abstract

Finite constructions of orders **684**, **1368**, and **5476** via Miyamoto.

- H_{684} = Paley-I ($q = 683$)
- H_{1368} = Paley-I ($q = 1367$) — preferred; also Double of H_{684}
- C_{1370} = Paley conference over GF(37^2), $q = 1369 = 37^2$
- H_{5476} = Miyamoto(H_{1368}, C_{1370})

SPK: 5476 Miyamoto-only bonus (NOT in SPK). Seeds 684/1368 enter SPK via Paley-I / Double.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_5476.py`

Results. overall_pass=true; units 63/63; wall 103.3562s; all max|HH −nI|=0.

Cutoff: 2026-09-15T20:43:28Z (~4:43PM ET)

Arithmetic

- Target $n = 5476 = 4 \cdot 1369$; $1369 = 37^2 \equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1368} and C_{1370} (Paley conference over prime-power 37^2).
- Seed preferred: Paley-I $q = 1367$ (prime $\equiv 3 \pmod 4$).
- Alternate: Double Paley-I half=684 with $q = 683$.
- Triage FAIL for 1368 Paley-II: $q = 683 \equiv 3 \pmod 4$.
- SPK: Paley-I $q = 5475$ **FAIL**; Paley-II $q = 2737$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only**.

Constructions (verified)

Object	Method	max|HH −nI|	Wall (s)
H_684	Paley-I q=683	0	0.4074
H_1368 Paley-I	Paley-I q=1367 (preferred)	0	2.251
H_1368 Double	Sylvester-double(H_684)	0	1.1112
C_1370	Paley conf GF(37^2) q=1369	0	1.5488
H_5476 Miyamoto	Miyamoto(H_1368, C_1370)	0	95.8336
H_5476 Paley-I	SKIP q=5475=3 · 1825	—	—
H_5476 Paley-II	SKIP q=2737 not pp	—	—

SPK closure note

Seeds include prior SPK ships through **5444** (Paley-I) + **340/1360** + verified **684/1368**; **exclude** Miyamoto-only bonuses including **5284** and **5476**. Known count 1171. Next constructive gap **2052** (OPEN).

Quantity	Value
5476 in SPK	**False** (Miyamoto-only)
5444 in SPK	**True** (prior Paley-I)
1368 / 684 in SPK	**True** (Paley-I / Double)
Next constructive gap	**2052** (OPEN)

Honesty

- clay_closed = false
- GATE-04 = LOCKED
- Hadamard conjecture = OPEN
- Gap 2052 = OPEN (method triage / non-finding ONLY)
- 5476_in_spk = false
- 5476_miyamoto_only = true

References (classical; no fake DOIs)

- Paley type I/II constructions (prime / prime-power fields).
- Symmetric Paley conference matrices for $q \equiv 1 \pmod 4$ (incl. GF(p^2)).
- Miyamoto construction of Hadamard matrices of order $4q$ from H_{q-1} + conference of order $q+1$.
- SPK = Sylvester Paley I/II Kronecker; Miyamoto not folded into SPK.